Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

If ${}^nC_4, {}^n\mathrm{C}_5$ and ${}^nC_6$ are in A.P., then a value of n can be

(A) 6

(B) 7

(C) 8

(D) 9

Q.2

Kunal Gaba has n objects, each of weight w. He weighs them in pairs and finds the sum of the weights of all possible pairs is $120\ \mathrm{g}$. When his friend Rakshit weighs them in triplets, the sum of all possible weights is $240\ \mathrm{g}$. The value of n is

(A) 7

(B) 6

(C) 5

(D) 10

Q.3

If $a_n = \sum_{r=0}^{n} \frac{1}{{}^nC_r}$, then $\sum_{r=0}^{n} \frac{r}{{}^nC_r}$ equals:

(A) $\frac{n}{2}a_n$

(B) $\frac{n}{4}a_r$

(C) na_n

(D) $(n-1)a_n$

Q.4

The sum $\sum_{i=0}^{m} \begin{pmatrix} 10 \\ i \end{pmatrix} \begin{pmatrix} 20 \\ m-i \end{pmatrix}$ (where $\begin{pmatrix} p \\ q \end{pmatrix} = 0$ if $p < q$) is maximum where m is

(A) 5

(B) 10

(C) 15

@IITJEE_Advanced

(D) 20

Q.5

Let T_n denote the number of triangles which can be formed by using the vertices of a regular polygon of n sides. If $T_{n+1} - T_n = 21$, then n equals

(A) 5

(B) 7

(C) 6

(D) 4

Q.6

A man wants to distribute 101 coins of a rupee each, among his 3 sons with the condition that no one receives more money than the combined total of the other two. The number of ways of doing this is equal to

(A) ${}^{103}C_2 - 3^{52}C_2$

(B) $\frac{{}^{103}C_2}{3}$

(C) $\frac{{}^{103}C_2}{6}$

(D) ${}^{103}C_2 - 3^{50}C_3$

Q.7

Four couples (husband and wife) decide to form
a committee of four members. Find the number of different committees that can be formed in which no couple finds a place.

(A) 12

(B) 14

(C) 16

(D) 24

Q.8

The sides $\mathrm{AB}, \mathrm{BC}, \mathrm{CA}$ of a traingle ABC have 3 , 4 and 5 interior points respectively on them. The total number of triangles that can be constructed by using these points as vertices is

(A) 220

(B) 204

(C) 205

@IITJEE_Advanced

(D) 195

Q.9

A committee of 4 persons is to be formed from 2 ladies, 2 old men and 4 young men such that it includes at least 1 lady, at least 1 old man and at most 2 young men. Then the total number of ways in which this committee can be formed is :

(A) 40

(B) 41

(C) 16

(D) 32

Q.10

The value of ' n ' for which
${}^{n-1}C_4 - {}^{n-1}C_3 - \frac{5}{4} \cdot {}^{n-2}P_2 < 0$, where $n \in N$

(A) $\{5, 6, 7, 8, 9, 10\}$

(B) $\{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$

(C) $\{1, 4, 5, 6, 7, 8, 9, 10\}$

(D) $(-\infty, 2) \cup (3, 11)$

Q.11

A student is to answer 10 out of 13 questions in an examination such that he must choose at least 4 from the first five questions. The number of choices available to him is

(A) 346

(B) 140

(C) 196

(D) 280

Q.12

There are 10 bags $\mathrm{B}_1, \mathrm{B}_2, \mathrm{B}_3, \ldots, \mathrm{B}_{10}$, which
contain $21, 22, \ldots, 30$ different articles respectively. The total number of ways to bring out 10 articles from a bag is

(A) ${}^{31}\mathrm{C}_{20} - {}^{21}\mathrm{C}_{10}$

(B) ${}^{31}\mathrm{C}_{21}$

(C) ${}^{31}\mathrm{C}_{20}$

@IITJEE_Advanced

(D) None of these

Q.13

The number of ways in which 10 boys can be divided into 2 groups of 5, such that two tallest boys are in two different groups, is equal to

(A) 70

(B) 35

(C) 252

(D) 126

Q.14

In a small village, there are 87 families, of which 52 families have at most 2 children. In a rural development programme 20 families are to be chosen for assistance, of which at least 18 families must have at most 2 children. In how many ways can the choice be made?

(A) ${}^{52}\mathrm{C}_{18}{}^{35}\mathrm{C}_{2}$

(B) ${}^{52}\mathrm{C}_{18} \times {}^{35}\mathrm{C}_{2} + {}^{52}\mathrm{C}_{19} \times {}^{35}\mathrm{C}_{1} + {}^{52}\mathrm{C}_{20}$

(C) ${}^{52}\mathrm{C}_{18} + {}^{35}\mathrm{C}_{2} + {}^{52}\mathrm{C}_{19}$

(D) ${}^{52}\mathrm{C}_{18}^{18} \times {}^{35}\mathrm{C}_{2} + {}^{35}\mathrm{C}_{1} \times {}^{52}\mathrm{C}_{19}$

Q.15

There are 3 oranges, 5 apples and 6 mangoes in a fruit basket. Number of ways in which at least one fruit can be selected from the basket is

(A) 168

(B) 167

(C) 125

(D) 124

Q.16

Out of 6 boys and 4 girls, a group of 7 is to be formed. In how many ways can this be done, if the group is to have a majority of boys?

(A) 120

(B) 80

(C) 90

(D) 100

@IITJEE_Advanced

Q.17

A seven digit number divisible by 9 is to be formed by using 7 out of number $\{1, 2, 3, 4, 5, 6, 7, 8, 9\}$. The number of ways in which this can be done is

(A) $7!$

(B) $2 \times 7!$

(C) $3 \times 7!$

(D) $4 \times 7!$

Q.18

If a denotes the number of permutations of $x + 2$ things taken all at a time, b the number of permutations of x things taken 11 at a time and c the number of permutations of $x - 11$ things taken all at a time such that $a = 182bc$, then the value of x will be

(A) 12

(B) 10

(C) 8

(D) 6

Q.19

If $\frac{2}{9!} + \frac{2}{3!7!} + \frac{1}{5!5!} = \frac{2^a}{b!}$, where $a, b \in \mathrm{N}$, then the ordered pair (a, b) is

(A) $(9, 10)$

(B) $(10, 9)$

(C) $(7, 10)$

(D) $(10, 7)$

Q.20

The tens digit of $1! + 2! + 3! + \ldots\ldots + 49!]$ is

(A) 1

(B) 2

(C) 3

(D) 4

Answers & Solutions

Q.1 Answer:

@IITJEE_Advanced

7

Solution:

As ${}^nC_4, {}^nC_5$ and ${}^nC_6$ are in A.P.,

$$2\left({}^nC_5\right) = {}^nC_4 + {}^nC_6$$

$$\Rightarrow \quad 2 = \frac{{}^nC_4}{{}^nC_5} + \frac{{}^nC_6}{{}^nC_5}$$

$$= \frac{n!}{4!(n-4)!}\frac{5!(n-5)!}{n!} + \frac{n!}{6!(n-6)!}\frac{5!(n-5)!}{n!}$$

$$\Rightarrow \quad 2 = \frac{5}{n-4} + \frac{n-5}{6} \Rightarrow n^2 - 21n + 98 = 0$$

$$\Rightarrow \quad n = 7, 14.$$

Q.2 Answer:

6

Solution:

According to the given condition $\left({}^nC_2\right)(2w) = 120 \Rightarrow n(n-1)w = 120$

and $\left({}^nC_3\right)(3w) = 240 \quad \Rightarrow \quad n(n-1)(n-2)w = 480$

Thus, $\frac{n(n-1)(n-2)w}{n(n-1)w} = \frac{480}{120}$

$\Rightarrow \quad n-2 = 4 \quad \Rightarrow \quad n = 6$

Q.3 Answer:

$\frac{n}{2}a_n$

Solution:

Let $b_n = \sum_{r=0}^{n} \frac{r}{{}^nC_r} = \sum_{r=0}^{n} \frac{n-r}{{}^nC_{n-r}} = \sum_{r=0}^{n} \frac{n-r}{{}^nC_r}$

$\Rightarrow \quad 2b_n = \sum_{r=0}^{n} \frac{r+(n-r)}{{}^nC_r} = n\sum_{r=0}^{n} \frac{1}{{}^nC_r} = na_n$

$\Rightarrow \quad b_n = \frac{n}{2}a_n.$

Q.4 Answer:

15

Solution:

We have $\sum_{i=0}^{m} \begin{pmatrix} 10 \\ i \end{pmatrix} \begin{pmatrix} 20 \\ m-i \end{pmatrix}$

${}^{10+20}C_{i+m-i}$

${}^{30}C_m$

$m = 15$, is max

But ${}^{30}C_m$ is maximum for $m = 15$.

Q.5 Answer:

7

@IITJEE_Advanced

Solution:

The number of triangles that can be formed by using the vertices of a regular polygon is ${}^{n}C_3$**. That is,** $T_n = {}^{n}C_3$

Now, $T_{n+1} - T_n = 21$

$\Rightarrow \quad {}^{n+1}C_3 - {}^{n}C_3 = 21$

$\Rightarrow \quad {}^{n}C_2 + {}^{n}C_3 - {}^{n}C_3 = 21 \quad \left[\because {}^{n+1}C_r = {}^{n}C_{r-1} + {}^{n}C_r\right]$

$\Rightarrow \quad \frac{1}{2}n(n-1) = 21 \Rightarrow n = -6$ **or** 7

As n **is a positive integer,** $n = 7$.

Q.6 Answer:

${}^{103}C_2$ - $3^{52}C_2$

Solution:

Let, the amount received by the sons be Rs. x, **Rs.** y, **and Rs** z **respectively, then**

$x \leq y + z = 101 - x \Rightarrow 2x \leq 101$.

$\Rightarrow x \leq 50, \quad y \leq 50, \quad z \leq 50$.

$x + y + z = 101$

So, the required number of ways is the coefficient of x^{101} **in** $1 + x + x^2 + \ldots + x^{51}{}^3$

or coefficient of x^{101} **in the expansion of** $\frac{1 - x^{50}{}^3}{1 - x}$

or coefficient of x^{101} **in the expansion of** $1 - x^{51}{}^3 1 - x^{-3}$

or coefficient of x^{101} **in the expansion of** $1 - {}^{3}C_1 x^{51} + \ldots . 1 - x^{-3}$

or coefficient of x^{101} **in the expansion of** $1 - x^{-3} - {}^{3}C_1 x^{51} 1 - x^{-3} = 1 \times {}^{103}C_{101} - 3^{52}C_{50} = {}^{103}C_2 - 3^{52}C_2$

Q.7 Answer:

16

Solution:

The number of committees of 4 gentlemen

$$= {}^{4}\mathrm{C}_4 = 1$$

@IITJEE_Advanced

The number of committees of 3 gentlemen, 1 wife

$$= {}^{4}C_{3} \times {}^{1}C_{1}$$

($\because$ after selecting 3 gentlemen only 1 wife is left who can be included) The number of committees of 2 gentlemen, 2 wives

$$= {}^{4}\mathrm{C}_{2} \times {}^{2}\mathrm{C}_{2}$$

The number of committees of 1 gentleman, 3 wives

$$= {}^{4}\mathrm{C}_{1} \times {}^{3}\mathrm{C}_{3}$$

The number of committees of 4 wives $= 1$ $\therefore$ The required number of committees

$$= 1 + 4 + 6 + 4 + 1 = 16$$

Q.8 Answer:

205

Solution:

We have in all 12 points. Since, 3 points are used to form a traingle, therefore the total number of traingles including the triangles formed by collinear points on AB, BC and CA is ${}^{12}C_{3} = 220$. But this includes the following :

The number of traingles formed by 3 points on $\mathrm{AB} = {}^{3}\mathrm{C}_{3} = 1$

The number of triangles formed by 4 points on $\mathrm{BC} = {}^{4}\mathrm{C}_{3} = 4$

The number of triangles fromed by 5 points on $\mathrm{CA} = {}^{5}\mathrm{C}_{3} = 10.$

Hence, required number of traingles $= 220 - (10 + 4 + 1) = 205.$

Q.9 Answer:

41

Solution:

@IITJEE_Advanced

L	O	Y
2	2	4
≥ 1	≥ 1	$2 \leq$

$\Rightarrow$

L	O	Y
1	1	2
1	2	1
2	1	1
2	2	0

Required number of ways

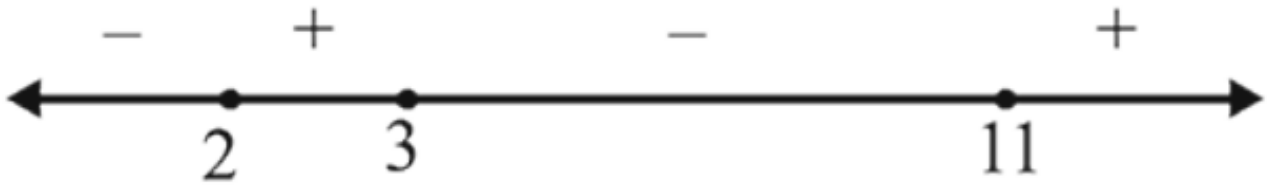
$= {}^2C_1 \times {}^2C_1 \times {}^4C_2 + {}^2C_1 \times {}^2C_2 \times {}^4C_1$
$\quad + {}^2C_2 \times {}^2C_1 \times {}^4C_1 + {}^2C_2 \times {}^2C_2 \times {}^4C_0$
$= 2 \times 2 \times \frac{4\times 3}{2} + 2 \times 1 \times 4 + 1 \times 2 \times 4 + 1 \times 1 \times 1$
$= 24 + 8 + 8 + 1 = 41$

Q.10 Answer:

$\{5, 6, 7, 8, 9, 10\}$

Solution:

We have, ${}^{n-1}C_4 - {}^{n-1}C_3 - \frac{5}{4} \cdot {}^{n-2}P_2 < 0$
$\Rightarrow \frac{(n-1)(n-2)(n-3)(n-4)}{4!} - \frac{(n-1)(n-2)(n-3)}{3!}$
$-\frac{5}{4}(n-2)(n-3) < 0$
$\Rightarrow (n-2)(n-3)(n-11)(n+2) < 0$
$\Rightarrow (n-2)(n-3)(n-11) < 0$
$[\because n + 2 > 0$ **for** $n \in N]$
$\Rightarrow n \in (-\infty, 2) \cup (3, 11)$



$\Rightarrow n \in (0, 2) \cup (3, 11)$
$\Rightarrow \mathrm{n} = 1, 4, 5, 6, 7, 8, 9, 10$
But ${}^{\mathrm{n}-1}\mathrm{C}_4$ **and** ${}^{\mathrm{n}-2}\mathrm{P}_2$ **both are meaningful for** $n \geq 5$
Hence, $\mathrm{n} = 5, 6, 7, 8, 9, 10$

Q.11 Answer:

@IITJEE_Advanced

196

Solution:

As for given question two cases are possible.

(i) Selecting 4 out of first five question and 6 out ofremaining 8 question $= {}^{5}C_{4} \times {}^{8}C_{6} = 140$ **choices.**

(ii) Selecting 5 out of first five question and 5 out of remaining 8 questions $= {}^{5}C_{5} \times {}^{8}C_{5} = 56$ **choices Therefore, total number of choices** $= 140 + 56 = 196.$

Q.12 Answer:

$^{31}\mathrm{C}_{20} - {}^{21}\mathrm{C}_{10}$

Solution:

The required number of ways $= {}^{21}\mathrm{C}_{10} + {}^{22}\mathrm{C}_{10} + {}^{23}\mathrm{C}_{10} + \ldots. + {}^{30}\mathrm{C}_{10}$

$= \left({}^{21}\mathrm{C}_{10} + {}^{21}\mathrm{C}_{11} + {}^{22}\mathrm{C}_{12} + {}^{23}\mathrm{C}_{13} + \ldots\right.$

$\left. + {}^{30}\mathrm{C}_{20}\right) - {}^{21}\mathrm{C}_{11}$

${}^{22}\mathrm{C}_{11} + {}^{22}\mathrm{C}_{12} + {}^{23}\mathrm{C}_{13} + \ldots.. + {}^{30}\mathrm{C}_{20} - {}^{21}\mathrm{C}_{10}$

$\left(\because {}^{\mathrm{n}}\mathrm{C}_{\mathrm{r}-1} + {}^{\mathrm{n}}\mathrm{C}_{\mathrm{r}} = {}^{\mathrm{n}+1}\mathrm{C}_{\mathrm{r}}\right)$

$= {}^{31}C_{20} - {}^{21}C_{10}$

Q.13 Answer:

70

Solution:

Excluding the 2 **tallest boys, we can divide the** 8 **boys into** 2

groups of 4 **each by**

$\frac{8!}{4!4!2!}$ **ways.**

The tallest boys can be assigned to the groups in 2 **ways.**

The desired number of ways are

$\frac{8!}{4!4!2!} \times 2! = \frac{8 \times 7 \times 6 \times 5}{4 \times 3 \times 2} = 70$

Q.14 Answer:

$^{52}\mathrm{C}_{18} \times {}^{35}\mathrm{C}_{2} + {}^{52}\mathrm{C}_{19} \times {}^{35}\mathrm{C}_{1} + {}^{52}\mathrm{C}_{20}$

Solution:

The following are the number of possible choices:

$^{52}\mathrm{C}_{18} \times {}^{35}\mathrm{C}_{2}$**(**$18$ **families having atmost 2 children and 2 selected from other type of families)**

$^{52}\mathrm{C}_{19} \times {}^{35}\mathrm{C}_{1}$ **(19 families having atmost 2**

children and 1 selected from other type of families) $^{52}\mathrm{C}_{20}$ **(All selected 20 families having atmost 2 children). Hence, the total number of possible choices is :=** $^{52}\mathrm{C}_{18} \times {}^{35}\mathrm{C}_{2} + {}^{52}\mathrm{C}_{19} \times {}^{35}\mathrm{C}_{1} + {}^{52}\mathrm{C}_{20}$

@IITJEE_Advanced

Q.15 Answer:

167

Solution:

Since each type of fruit would be considered identical.

Hence, the number of ways in which at least one fruit can be selected from the basket

$= 3 + 1\ 5 + 1\ 6 + 1\ -1 = 167$

Q.16 Answer:

100

Solution:

The boys are in majority, if the groups are $4B3G, 5B2G, 6B \pm G$

Total number of combination

$$= {}^{6}c_4 \times {}^{4}c_3 + {}^{6}c_5 \times {}^{4}c_2 + {}^{6}c_6 \times {}^{4}c_1$$
$$= 15 \times 4 + 6 \times 6 + 1 \times 4$$
$$= 60 + 36 + 4 = 100$$

Sotal number of combinction

$= {}^{6}c_4 \times {}^{4}c_3 + {}^{6}c_5 \times {}^{4}c_2 + {}^{6}c_6 \times {}^{4}c_1$

$= 15 \times 4 + 6 \times 6 + 1 \times 4$

$= 60 + 36 + 4 = 100$

Hence option (D) is correet.

Q.17 Answer:

$4 \times 7!$

Solution:

Sum of 7 digits = a multiple of 9 Since sum of number $1, 2, 3, \ldots\ldots, 8, 9$ **is 45 (Since a number is divisible by 9 if sum of its digits is divisible by 9 .) So, two left number should also have sum as 9. The pairs to be left are** $(1, 8), (2, 7), (3, 6), (4, 5)$**. With each pair left, number of 7 digit numbers** $= 7!$**. So, with all 4 pairs total seven digits number** $= 4 \times 7!$

Q.18 Answer:

12

Solution:

${}^{x+2}\mathrm{P}_{x+2} = a \Rightarrow a = (x+2)!$

${}^{x}\mathrm{P}_{11} = b \Rightarrow b = \frac{x!}{(x-11)!}$ **s and** ${}^{x-11}\mathrm{P}_{x-11} = c$

$\Rightarrow \quad c = (x-11)!; a = 182bc$

$(x+2)! = 182\frac{x!}{(x-11)!}(x-11)!$

$\Rightarrow \quad (x+2)(x+1) = 182 = 14 \times 13$

$x + 1 = 13 \quad \therefore x = 12$

@IITJEE_Advanced

Q.19 Answer:

$(9, 10)$

Solution:

$\frac{2}{9!}+\frac{2}{3!7!}+\frac{1}{5!5!}$
$=\frac{1}{1!9!}+\frac{1}{3!7!}+\frac{1}{5!5!}+\frac{1}{3!7!}+\frac{1}{9!1!}$
$=\frac{1}{10!}\left\{{}^{10}\mathrm{C}_1+{}^{10}\mathrm{C}_3+{}^{10}\mathrm{C}_5+{}^{10}\mathrm{C}_7+{}^{10}\mathrm{C}_9\right\}$
$=\frac{1}{10!}\left(2^{10-1}\right)=\frac{2^9}{10!}=\frac{2^a}{b!}$ **(given)**
$\Rightarrow \quad a=9, b=10$

Q.20 Answer:

1

Solution:

We have $1!+2!+3!+4!=33$.
Also $5!=120, 6!=720, 7!=5040, 8!=40320$ **and** $9!=326880$.
Thus, the tens digit of $1!+2!+\ldots+9!$ **is 1 .**
Also, note that $n!$ **is divisible by 100 for all** $n \geq 10$,
so that tens digits of $10!+11!+\ldots .+49!$ **is zero.**
Therefore, tens digit of $1!+2!+\ldots .+49!$ **is 1**

@IITJEE_Advanced